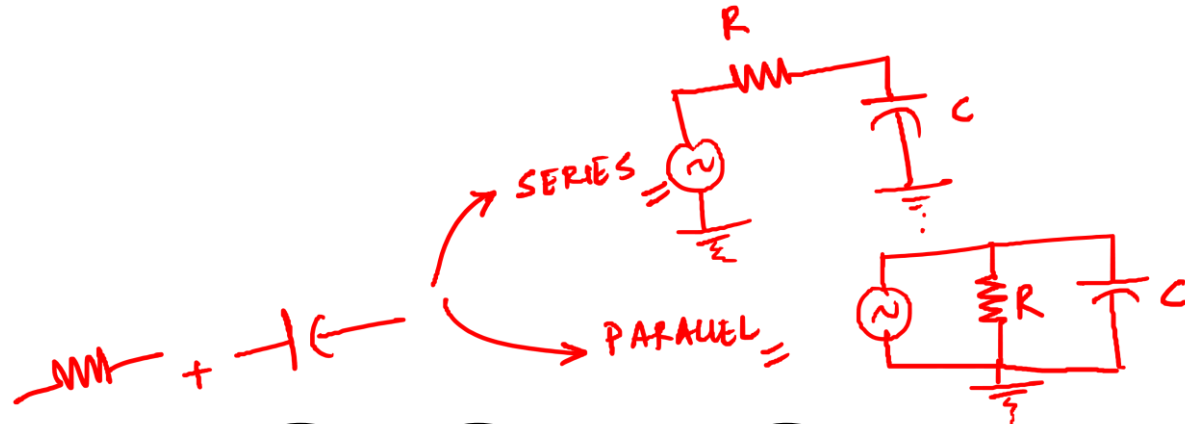




RC CIRCUIT

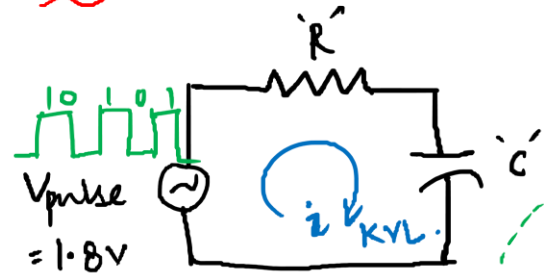
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DERIVATION OF CHARGING & DISCHARGING EQUATIONS OF A CAPACITOR

Consider a R-C ckt; with Resistance 'R' and Capacitance 'C'.



Putting KVL: We get

$$V_{\text{pulse}} - V_R - V_C = 0$$

$$V_{\text{pulse}} = V_R + V_C$$

$$V_{\text{pulse}} = i(t) \cdot R + \frac{q(t)}{C}$$

$$V_{\text{pulse}} = \left(\frac{dq}{dt} \right) R + \frac{q(t)}{C}$$

$V_C(t)$ → Charge ✓
→ Discharge ✓

$$i = \frac{dV_C(t)}{dt}$$

$$Q = CV$$

$$V = \frac{Q}{C}$$

$$i = \frac{dq}{dt}$$

Comparing it with its General form; We get

$$\frac{dy}{dx} + P(x).y = Q(x)$$

$$\left. \begin{array}{l} x \rightarrow t \\ y \rightarrow q \end{array} \right\} \begin{array}{l} P(x) = \frac{1}{R \cdot C} \\ Q(x) = \frac{V_{\text{pulse}}}{R} \end{array}$$

$$\left\{ \frac{dq}{dt} + \frac{q(t)}{R \cdot C} = \frac{V_{\text{pulse}}}{R} \right\}$$

$$\text{I.F.} \rightarrow e^{\int P(x).dx} = e^{t/R \cdot C}$$

(Integrating Factor)

①

We get;

$$\text{I} \rightarrow \frac{d}{dt} \left[q \cdot e^{t/R \cdot C} \right]$$

$$\text{II} \rightarrow \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C}$$

Now, Dividing 'R' throughout the eqn; We get

$$\frac{V_{\text{pulse}}}{R} = R \cdot \frac{dq}{dt} \left(\frac{1}{R} \right) + \frac{q(t)}{C} \left(\frac{1}{R} \right)$$

Now, Multiplying the Entire eqn By the Integrating Factor i.e. →

$$e^{t/R \cdot C} \times \left\{ \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \right\} = e^{t/R \cdot C} \times \frac{V_{\text{pulse}}}{R}$$

I

II

$$\Rightarrow \frac{V_{\text{pulse}}}{R} = \left(\frac{dq}{dt} \right) + \frac{q(t)}{R \cdot C}$$

→ highest order = 1

Hence; it's a FIRST ORDER DIFFERENTIAL eqn !!

$$\Rightarrow q(t) \cdot e^{t/R \cdot C} = \int \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C} dt = \underbrace{C \cdot V_{\text{pulse}} \cdot e^{t/R \cdot C}}_{\text{Capacitance}} + \underbrace{K}_{\text{Arbitrary Integration Constant} \dots}$$

$$\text{@ time 't' = 0 :- } \int \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C} dt \Big|_0^0 = 0 = C \cdot V_{\text{pulse}} \cdot e^{t/R \cdot C} + K$$

$$= \boxed{C \cdot V_{\text{pulse}} + K} \Rightarrow K = -C \cdot V_{\text{pulse}}$$

$$\text{Similarly; @ time 't' = } C \cdot V_{\text{pulse}} (1 - e^{-t/R \cdot C}) = q(t).$$

{ $\because$ Since; $Q = CV$

$$q(t) = C \cdot V(t)$$

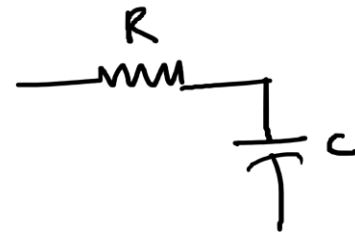
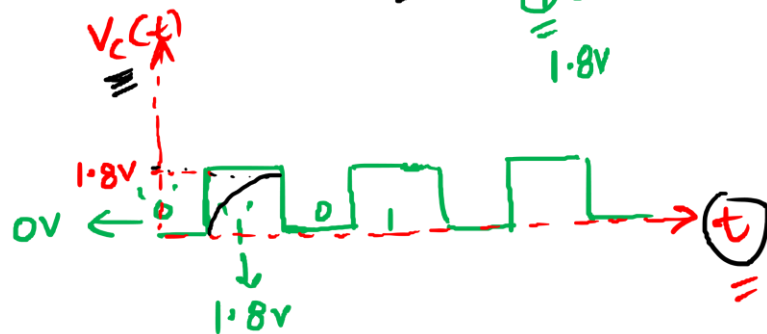
$$\downarrow$$

$$V_{\text{pulse}} (1 - e^{-t/R \cdot C}) =$$

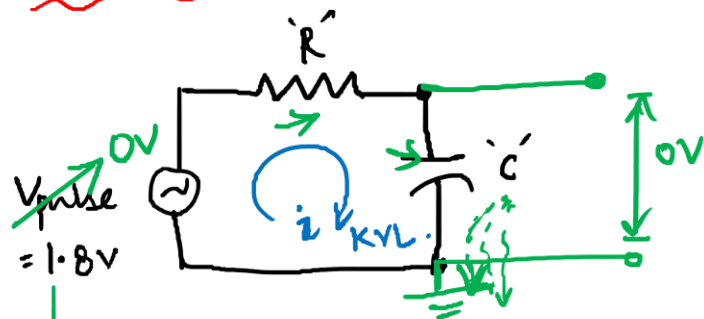
$$\text{So; } V_C(t) = \frac{q(t)}{C} = \frac{q}{C} \cdot V_{\text{pulse}} (1 - e^{-t/R \cdot C}) \Rightarrow \boxed{V_C(t) = V_{\text{pulse}} (1 - e^{-t/R \cdot C})}$$

Unknown

① 01010
= 1.8V



Consider a R-C ckt; with Resistance 'R' and Capacitance 'C'.



Putting KVL: We get

$$V_R - V_C = 0 \quad \checkmark$$

$$0 = V_R + V_C \quad \checkmark \quad \boxed{V_R = V_C}$$

$$0 = i(t) \cdot R \pm \frac{q(t)}{C}$$

$$0 = \frac{dq}{dt} \cdot R \pm \frac{q(t)}{C}$$

Now; dividing 'R' throughout the eqn; We get-

$$\frac{V_{\text{pulse}}}{R} = \frac{dq}{dt} \left(\frac{1}{R} \right) + \frac{q(t)}{C} \left(\frac{1}{R} \right) \quad \checkmark$$

$$\Rightarrow 0 = \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \rightarrow \text{highest order} = 1$$

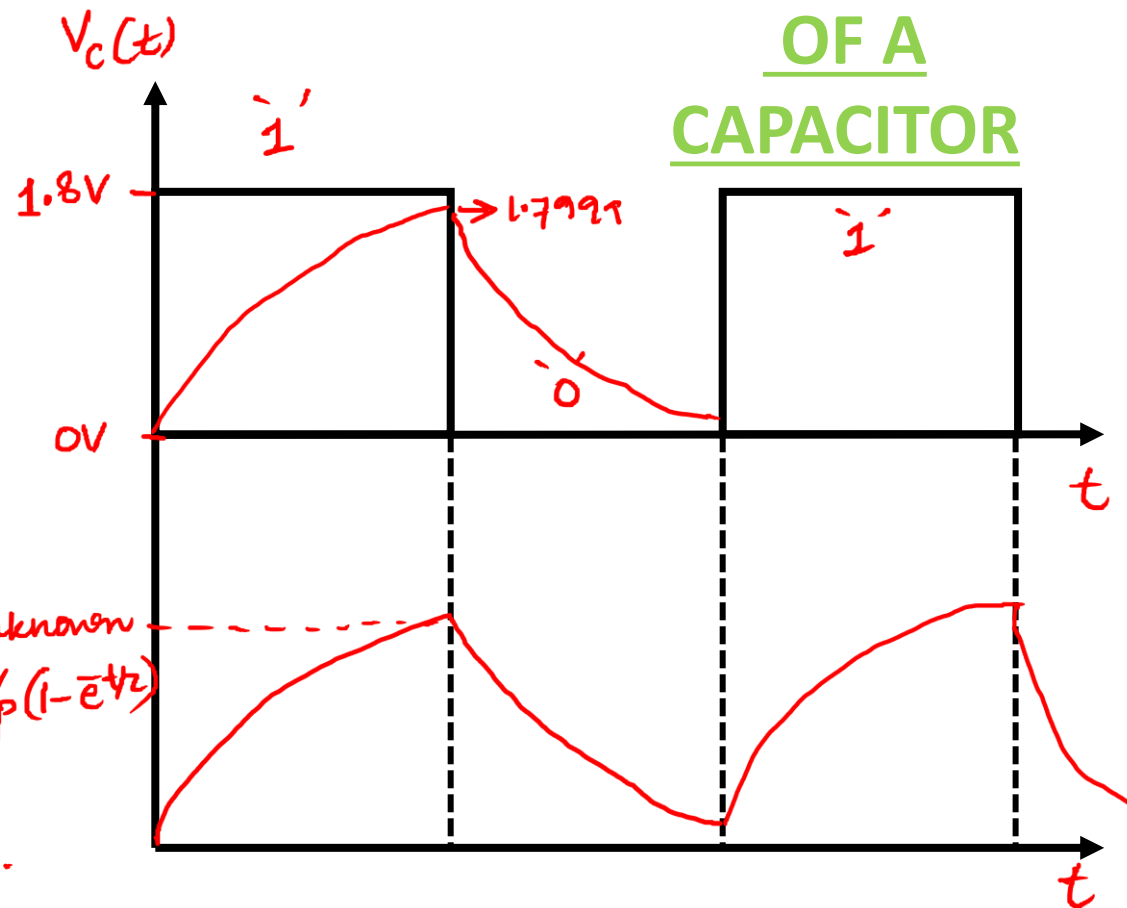
Hence; it's a FIRST ORDER DIFFERENTIAL EQN !!

$$\Rightarrow \frac{dq}{dt} = -\frac{q}{R \cdot C} \quad \checkmark \Rightarrow q(t) = q_0 \cdot e^{-t/R \cdot C}$$

Similarly;

$$V_C(t) \text{ DISCHARGING} = V_{\text{pulse}} \times e^{-t/R \cdot C}$$

DISCHARGING EQUATION OF A CAPACITOR



What is 1Z 57 & 10Z

unit $\rightarrow$ seconds (S) $\uparrow$ time $\uparrow$ slow $\uparrow$ Fast $\downarrow$
 (S) $\downarrow$ time $\downarrow$ slow $\downarrow$ Fast $\uparrow$

$[t_{ao}] \rightarrow$ are the Measures of HOW QUICKLY; A capacitor CHARGES and DISCHARGES...



injection to cut
 [NOT PREPARED] SLOW CHARGING & DISCHARGING =

MEDIUM SPEED FAST CHARGING & DISCHARGING =

Considering 1.8V is the Supply voltage. NOTE: More the speed of CHARGING & DISCHARGING, Less the Voltage $V_c(t)$.

$$V_1 > V_2 > V_3$$

$$1.8V = 1.799V = 1.15V = 0.6V$$

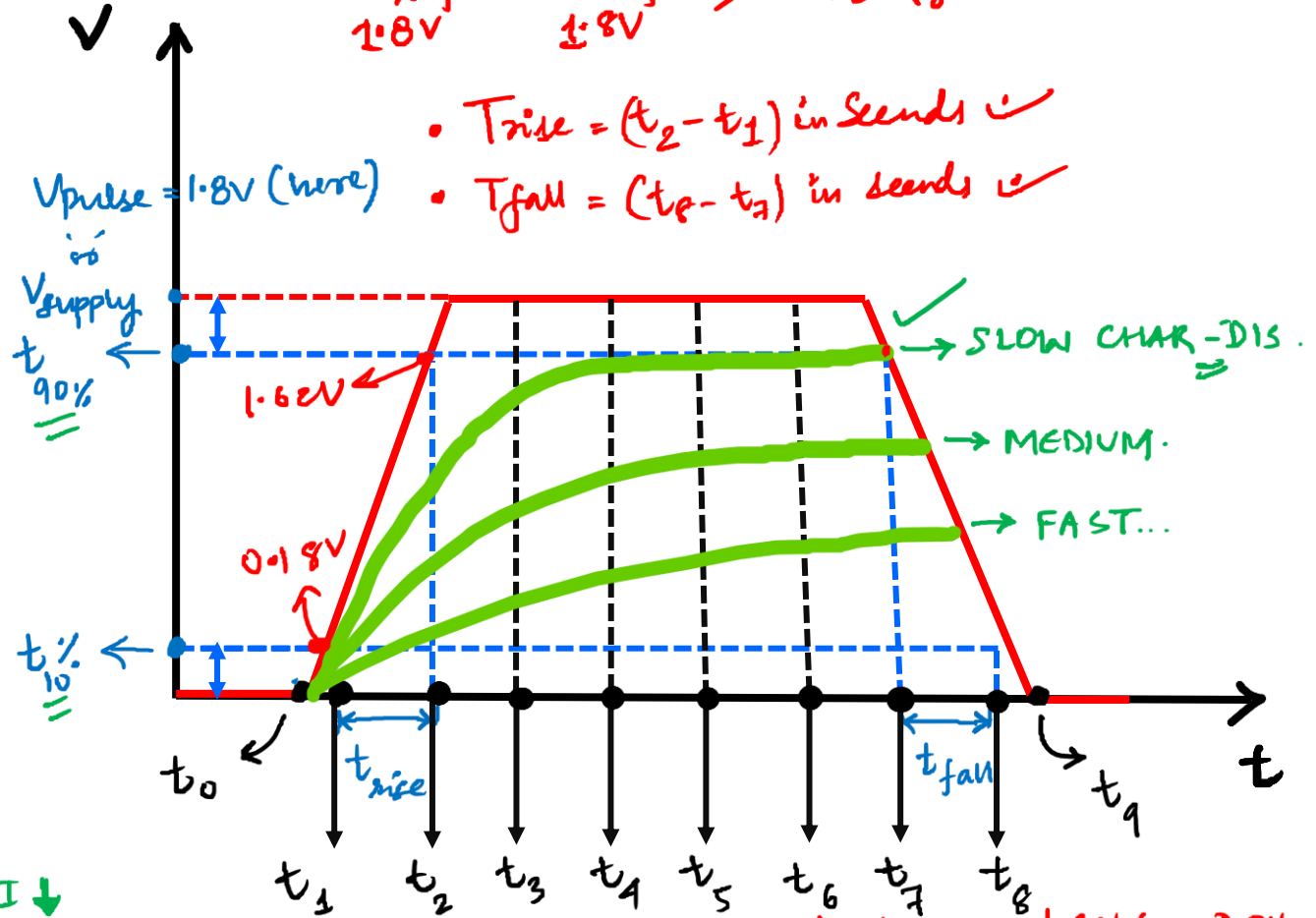
$$P = V \cdot I$$

$$V \downarrow P \downarrow \rightarrow I \downarrow$$

Remember: A CAPACITOR NEVER FULLY CHARGES TO V_{dd} or FULLY DISCHARGES TO GND .

$T_{rise} = t_{10\%}$ to $t_{90\%}$ of corresponding V_{supply} .

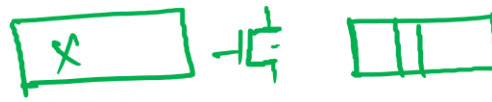
$$10\% \text{ of } 1.8V - 90\% \text{ of } 1.8V \rightarrow 0.18V \text{ to } 1.62V$$



- $T_{rise} = (t_2 - t_1)$ in seconds
- $T_{fall} = (t_8 - t_7)$ in seconds

$$A_v = \frac{V_{out}}{V_{in}} = 1$$

$$V_{out} = V_{in} = 1.8V/2 = 0.9V = 900mV$$



Charging and Discharging Percentage vs. Time Constants (τ)

Time (t)	Charging $V(t) = V_f(1 - e^{-t/\tau})$	% Charged	Discharging $V(t) = V_i e^{-t/\tau}$	% Discharged
1τ	$V_f(1 - e^{-1}) = 0.632 V_f$	63.2%	$V_i e^{-1} = 0.368 V_i$	36.8%
2τ	$V_f(1 - e^{-2}) = 0.865 V_f$	86.5%	$V_i e^{-2} = 0.135 V_i$	13.5%
3τ	$V_f(1 - e^{-3}) = 0.950 V_f$	95.0%	$V_i e^{-3} = 0.0498 V_i$	4.98%
4τ	$V_f(1 - e^{-4}) = 0.982 V_f$	98.2%	$V_i e^{-4} = 0.0183 V_i$	1.83%
5τ	$V_f(1 - e^{-5}) = 0.993 V_f$	99.3%	$V_i e^{-5} = 0.0067 V_i$	0.67%
6τ	$V_f(1 - e^{-6}) = 0.9975 V_f$	99.75%	$V_i e^{-6} = 0.0025 V_i$	0.25%
7τ	$V_f(1 - e^{-7}) = 0.9991 V_f$	99.91%	$V_i e^{-7} = 0.00091 V_i$	0.091%
8τ	$V_f(1 - e^{-8}) = 0.99966 V_f$	99.966%	$V_i e^{-8} = 0.00034 V_i$	0.034%
9τ	$V_f(1 - e^{-9}) = 0.99988 V_f$	99.988%	$V_i e^{-9} = 0.00012 V_i$	0.012%
10τ	$V_f(1 - e^{-10}) = 0.99995 V_f$	99.995%	$V_i e^{-10} = 0.000045 V_i$	0.0045%

